

Employee REST API Project Report (Revised v2)

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1. Executive Summary & Architecture Overview

This submission delivers an enterprise Employee REST API built with Spring Boot and SQLite. The system implements complete CRUD persistence, service isolation, and relational mappings.

2. CRUD Operations & REST Endpoints

- GET /api/v1/employees - Retrieves all active employee records.
- GET /api/v1/employees/{id} - Fetches detailed record by ID.
- POST /api/v1/employees - Creates employee with department and salary fields.
- PUT /api/v1/employees/{id} - Updates employee details.
- DELETE /api/v1/employees/{id} - Soft deletes employee record.

3. Database Integration & Entity Schema

Configured Spring Data JPA with @Entity Employee and JpaRepository<Employee, Long>. Includes primary key generation, indexed email column, and SQL schema migrations.

4. Input Validation & Request Constraints

Implemented Jakarta Validation with @Valid, @NotNull, @NotBlank, and @Email on EmployeeDTO. Ensures salary > 0, email follows standard RFC regex, and department name is mandatory.

5. Centralized Error & Exception Handling

Added @ControllerAdvice GlobalExceptionHandler catching ResourceNotFoundException and MethodArgumentNotValidException, returning standardized JSON ErrorResponse with timestamp.

6. Documentation & Setup Instructions

Complete README provided with curl commands, Swagger/OpenAPI endpoint mappings, and local execution commands: mvn spring-boot:run.